CIE XYZ colour space
One of the first mathematical descriptions of a colour space, based on a three-colour additive model, this was devised in 1931 by the International Commission on Illumination (CIE)

Why Red-Green-Blue?
Our eyes have three different sensors for colour, They have peak sensitivities in the red, green and blue ranges of the visible spectrum

use the help system to learn how to use the device. This does not contain enough much information – not by a long way.

The software itself does not give the user much information either – we also found with the Spyder for calibrating monitors that the data derived by the system is not really accessible to the user. You can see what the developers are trying to do – make the system easy to use and not confuse things with too much information; but this is a device for enthusiastic amateurs and professionals – just the kind of people who are likely to want at least the ability to dig deeper under the hood.

The process of calibrating a printer contains many details, but it is simple to describe overall. You print out a test page that contains a large number of defined colour patches, and then each of these is read by the Spyder colorimeter. When all have been read, the software compares the expected colours with those that have been detected, and then proceeds to create a profile for use when printing. On the Windows XP system we used, the profile was placed in the windows\system32\spool\drivers\color\ directory. (For some bizarre reason, different version of Windows use other directories.) From there it is readily found by programs such as Photoshop. And now for the details.

One important issue with calibrating a printer is that if you can, turn off any colour management that the printer driver itself might otherwise be trying to do. It is not always easy to check this, and it is not always possible to turn it off. For trying out the Spyder, we used an HP Photosmart C8188 All-in-One. Its printer driver gives two options under colour management – Adobe RGB or ColorSmart/sRGB. We used Adobe RGB as this has the largest gamut.


The Spyder colorimeter set up for calibration. The laptop screen shows the same set of colour patches as on the test print that is being measured

In a recent document, HP has described that a new, third, option is being added to its printer drivers: “Managed by application”. This turns off printer driver-based colour management, and the printer will not make any colour conversions to the image data. This of course assumes that you will be using software that will be able to handle the colour management, using the profile that you create with the Spyder or other colorimeter. If you are going to buy a colour printer, make sure that the colour management can be turned off.

Needless to say, not just the driver colour management, but also any other printer driver settings need to be the same when you perform the calibration and create the profile as when you later use the printer. If you intend to use different paper and ink combinations, and want accurate colour with these, each combination will need to be profiled separately.

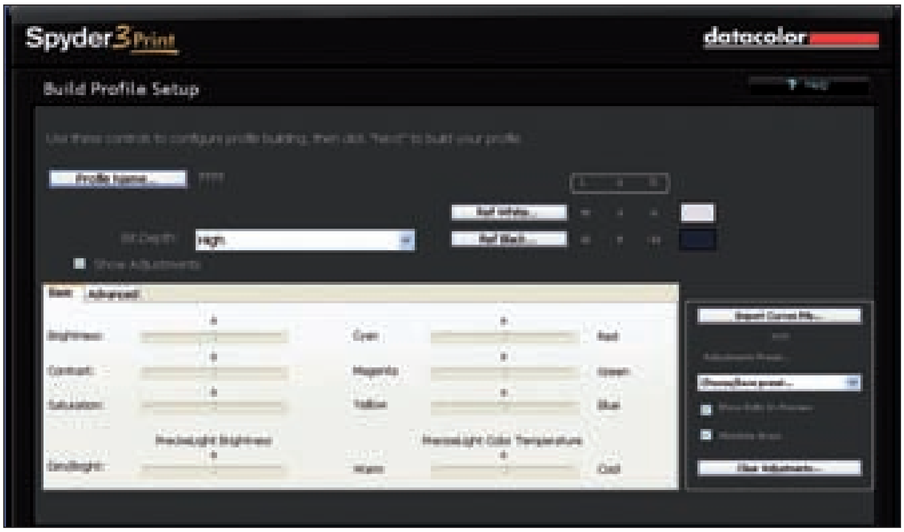
The next step is to run the Spyder software and calibrate the colorimeter. The small stand in which the Spyder can sit when not being used contains a small white patch under the Spyder sensor. When you start the software, you check that it can see the colorimeter, seated in its stand, and then select a menu option for calibration. Of course, it makes more sense to calibrate using the paper that will be used in printing. I have been told this will be particularly beneficial when there are high levels of optical brighteners in the paper, but it seems better to do this all the time.

After noting details regarding the printer and media to be used, you are lead through the process of checking the printer’s quality and media settings; you then come to print out the print sample that will be used for calibration. There are five from which you can choose:

Fast target – 150 patches
High quality target – 225 patches
Expert target – 729 patches (on three sheets, small paper)
Expert target – 729 patches (on three sheets, large paper)
Extended Greys – for black and white printing

For our testing, we used the fast target with 150 colour patches. Time did not allow us to compare the results from using the 150 patch calibration against the 225 or 729 patch methods. There are other factors coming later that perhaps might make a great difference.

When you perform the actual calibration,



The screen display of the test colour patches. The red triangle – fourth column, third row – is essentially a cursor indicating the current patch